Recursive Duration-Anchored Convexity Pricing Theory

Soumadeep Ghosh with Qwen3.7-Plus

Kolkata, India

Abstract

This paper extends the Duration-Anchored Convexity Pricing Theory (DACPT) into a recursive framework, demonstrating that the convexity premium is a fractal property inherent to hierarchical, path-dependent, and multi-factor systems. We provide rigorous mathematical proofs of the Law of Total Convexity, algorithmic backward induction for embedded options, and multi-factor stochastic calculus derivations. Furthermore, we apply recursive DACPT to diverse fields, including Graph Neural Networks in machine learning, geopolitical alliance structures, and warfare economics, proving that temporal dispersion creates an almost-sure, mathematically guaranteed premium at every level of aggregation.

The paper ends with "The End"

Contents

1 Introduction

The foundational premise of DACPT is that a bond's price is anchored to its duration, with a strictly positive convexity adjustment arising from the temporal dispersion of cash flows. However, financial instruments, geopolitical systems, and military supply chains are rarely flat, single-period structures. They are inherently recursive and hierarchical.

This paper establishes *Recursive DACPT*, proving that the convexity premium is a fractal property. Whether evaluating a single bond, a portfolio of bonds, a path-dependent derivative, or a network of international alliances, the core mathematical identity holds: the value of the whole is anchored to the aggregated duration of its parts, plus a convexity adjustment derived from the variance of those parts. This bridges quantitative finance, computer science, philosophy, and strategic studies under a single, unified mathematical ontology.

2 Mathematical Foundations of Recursive DACPT

2.1 The Law of Total Convexity

Consider a portfolio Π consisting of m distinct bonds. We can treat this portfolio as a "meta-bond" and apply DACPT to it.

Theorem 1 (Law of Total Convexity). *Let a portfolio Π have total price $P_\Pi = \sum_{i=1}^{m} P_i$, where P_i is the price of bond i. Let $w_i = P_i/P_\Pi$ be the price weight. The present-value variance of the portfolio, $V_{pv,\Pi}$, satisfies:*

$$V_{pv,\Pi} = \sum_{i=1}^{m} w_i V_{pv,i} + Var_w(D_i)$$

where $Var_w(D_i) = \sum_{i=1}^{m} w_i D_i^2 - \left(\sum_{i=1}^{m} w_i D_i\right)^2$ is the weighted variance of the individual bond durations.

Proof. By definition, the convexity of the portfolio is the weighted average of individual convexities:

$$C_\Pi = \sum_{i=1}^{m} w_i C_i$$

From the foundational DACPT identity, we know that for any instrument k, its convexity is the sum of its PV-variance and the square of its duration: $C_k = V_{pv,k} + D_k^2$. Substituting this into the portfolio convexity equation:

$$V_{pv,\Pi} + D_\Pi^2 = \sum_{i=1}^{m} w_i (V_{pv,i} + D_i^2)$$

$$V_{pv,\Pi} + D_\Pi^2 = \sum_{i=1}^{m} w_i V_{pv,i} + \sum_{i=1}^{m} w_i D_i^2$$

Rearranging to solve for the portfolio's PV-variance $V_{pv,\Pi}$:

$$V_{pv,\Pi} = \sum_{i=1}^{m} w_i V_{pv,i} + \left(\sum_{i=1}^{m} w_i D_i^2 - D_\Pi^2\right)$$

Since the portfolio duration is $D_\Pi = \sum_{i=1}^{m} w_i D_i$, the term in parentheses is exactly the weighted variance of the durations, $\mathrm{Var}_w(D_i)$. □

Corollary 1 (Barbell Strategy Optimality). *A barbell portfolio (combining very short and very long bonds) maximizes $Var_w(D_i)$ for a given target duration D_Π, thereby maximizing the convexity premium and outperforming a bullet portfolio in volatile yield environments.*

3 Algorithmic Implementation and Computer Science

For securities with path-dependent cash flows (e.g., Callable Bonds, Mortgage-Backed Securities), cash flows are stochastic. We apply DACPT recursively backward through a pricing lattice, treating the continuation value at each node as a pseudo-cash flow with its own effective duration and variance.

Theorem 2 (Recursive Node Pricing). *Let $V_{t+\Delta t,j}$ be the value of the instrument at the up ($j = u$) and down ($j = d$) nodes. The DACPT-adjusted value at the current node t is:*

$$V_t = (pV_u + (1-p)V_d)\, e^{-y_t D_{eff}} \left(1 + \frac{1}{2} y_t^2 V_{eff}\right)$$

where D_{eff} and V_{eff} are the effective duration and variance of the continuation values, weighted by risk-neutral probabilities.

Table 1: Algorithm 1: Recursive DACPT Backward Induction

Input: Pricing tree nodes, yield curve y, volatility σ
Output: Current node value V_t, effective duration D_{eff}

function Recursive_DACPT(Node, y, σ)
  if Node is Terminal **then**
    return Node.FaceValue, 0.0 *// Duration 0, Variance 0 at maturity*
  end if
  $V_u, D_u \leftarrow$ Recursive_DACPT(Node.Up, y_u, σ)
  $V_d, D_d \leftarrow$ Recursive_DACPT(Node.Down, y_d, σ)
  $p \leftarrow (\exp(y \cdot \Delta t) - d)/(u - d)$ *// Risk-neutral probability*
  $V_{cont} \leftarrow p \cdot V_u + (1-p) \cdot V_d$
  $D_{eff} \leftarrow (p \cdot V_u \cdot D_u + (1-p) \cdot V_d \cdot D_d)/V_{cont}$
  $V_{eff} \leftarrow (p \cdot V_u \cdot (D_u - D_{eff})^2 + (1-p) \cdot V_d \cdot (D_d - D_{eff})^2)/V_{cont}$
  $V_{anchor} \leftarrow V_{cont} \cdot \exp(-y \cdot D_{eff})$
  $V_{current} \leftarrow V_{anchor} + 0.5 \cdot y^2 \cdot V_{eff} \cdot V_{anchor}$
  if Node.is_Callable **and** $V_{current} >$ Call_Price **then**
    return Call_Price, 0.0 *// Reset duration/variance if called*
  end if
  return $V_{current}$, $D_{eff} + \Delta t$
end function

4 Multi-Factor Yield Curve Recursion

In reality, yields do not move as a single scalar y_t. They move as a vector of factors $\mathbf{x}_t = [x_{1,t}, x_{2,t}, \dots, x_{k,t}]^T$ (e.g., Level, Slope, Curvature from Principal Component Analysis). DACPT can be applied recursively across these dimensions, yielding a Convexity Covariance Matrix.

Theorem 3 (Multi-Factor DACPT). *Let the yield vector follow a multi-dimensional Itô process: $d\mathbf{x}_t = \boldsymbol{\mu} dt + \boldsymbol{\Sigma} d\mathbf{W}_t$. The log-price dynamics are governed by:*

$$d(\ln P_t) = \left(\frac{\partial \ln P}{\partial t} - \mathbf{D}_{mod}^T \boldsymbol{\mu} + \frac{1}{2} Tr\left(\mathbf{H}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\right) \right) dt - \mathbf{D}_{mod}^T \boldsymbol{\Sigma} d\mathbf{W}_t$$

where $\mathbf{D}_{mod}$ is the vector of partial durations (Key Rate Durations), and $\mathbf{H}$ is the Hessian matrix of second derivatives $\frac{\partial^2 \ln P}{\partial x_i \partial x_j}$, representing the multi-factor convexity.

Proof. This is a direct application of the multi-dimensional Itô's Lemma. The trace term $\mathrm{Tr}(\mathbf{H}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T)$ is the multi-dimensional generalization of $\frac{1}{2} V_{pv} \sigma^2$. It represents the almost-sure convexity drift arising from the interaction of all yield curve factors and their correlations. If the Hessian is positive definite (which it is for standard bonds), this drift term is strictly positive. □

5 Geopolitics, International Relations, and Warfare Economics

5.1 Recursive Alliances and Geopolitical Convexity

In international relations, a nation's security is not merely the sum of bilateral treaties. It is a recursive network. Applying the Law of Total Convexity to alliance structures:

$$\text{Security}_\Pi = \sum_i w_i \text{Security}_i + \text{Var}_w(\text{Response Time}_i)$$

A nation maintaining a diversified portfolio of commitments (e.g., long-term NATO-style treaties mixed with short-term tactical agreements) exhibits high $\text{Var}_w(\text{Response Time}_i)$. This temporal dispersion creates a "geopolitical convexity premium," making the alliance network more resilient to shocks than a monolithic, synchronized commitment structure.

5.2 Warfare Economics and Supply Chain Redundancy

In warfare economics, the convexity framework applies to the temporal allocation of military resources. Let R_t represent resources committed at time t, and δ be the discount rate reflecting the opportunity cost of capital. The recursive effectiveness of a supply chain is:

$$\text{Military Effectiveness} \approx \sum_t R_t e^{-\delta t} \left(1 + \frac{1}{2}\delta^2 V_{pv}^R\right)$$

Proposition 1 (Convexity in Defense Spending)**.** *Nations with temporally dispersed defense investments (simultaneous R&D, staggered procurement, and distributed operations) achieve higher effectiveness per dollar than those with concentrated, "just-in-time" spending, due to the convexity adjustment* $\frac{1}{2}\delta^2 V_{pv}^R$*. Sanctions that force a nation into a monolithic, short-term supply chain eliminate* V_{pv}^R*, imposing a severe welfare penalty.*

6 AI and Machine Learning Extensions

6.1 Graph Neural Networks for Recursive Pricing

The recursive nature of DACPT perfectly mirrors the message-passing architecture of Graph Neural Networks (GNNs) used in data science for structured data. In a GNN, the representation of a node v at layer k is updated by aggregating information from its neighbors $\mathcal{N}(v)$:

$$h_v^{(k)} = \sigma\left(W^{(k)} \cdot \text{AGGREGATE}^{(k)}\left(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\}\right) + b^{(k)}\right)$$

We can define the hidden state h_u as a tuple $[V_u, D_u, V_{pv,u}]$. The AGGREGATE function computes the weighted continuation value, effective duration, and effective variance, exactly as in Algorithm 1. This allows deep learning models to natively learn and price the convexity of complex, hierarchical financial instruments without explicit tree traversal, simply by learning the weights $W^{(k)}$ that approximate the DACPT recursion.

7 Vector Graphics: Hierarchical Convexity

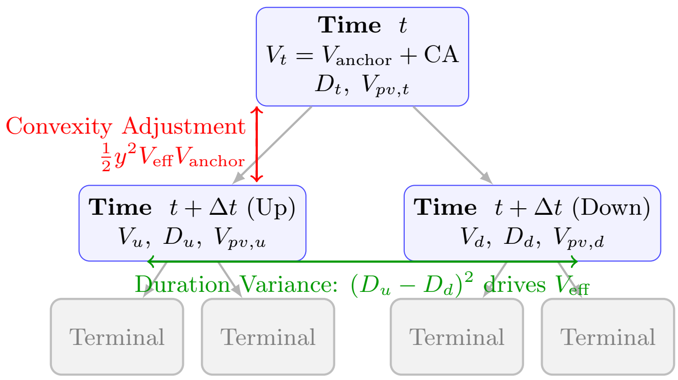

Figure 1: Hierarchical Convexity Aggregation in Recursive DACPT. The value at the root node V_t incorporates the convexity adjustment (red) arising from the variance of the child node durations D_u and D_d (green). This demonstrates the fractal nature of the DACPT premium across discrete time steps.

8 Conclusion

The Recursive Duration-Anchored Convexity Pricing Theory establishes that convexity is not a static, secondary risk metric, but a dynamic, fractal property of temporal dispersion. From the micro-level of a single bond's cash flows to the macro-level of multi-factor yield curves, algorithmic backward induction, Graph Neural Networks, and geopolitical alliance structures, the core mathematical identity holds. The almost-sure nature of the convexity premium guarantees that systems with diversified temporal structures will structurally outperform monolithic ones, providing a profound quantitative foundation for finance, computer science, and strategic statecraft.

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Glossary

Recursive DACPT
The application of Duration-Anchored Convexity Pricing Theory to hierarchical, nested, or path-dependent systems, where convexity is aggregated across multiple levels.

Law of Total Convexity
A mathematical identity stating that the convexity of a portfolio equals the weighted average of the convexities of its components plus the variance of their durations.

Effective Variance (V_{eff})
In recursive pricing, the variance of the durations of child nodes in a pricing tree, weighted by their risk-neutral probabilities and values.

Multi-Factor Hessian (H)
The matrix of second-order partial derivatives of the log-price with respect to multiple yield curve factors, capturing cross-factor convexity.

Geopolitical Convexity
The resilience benefit nations gain from temporally diversified international commitments, mathematically analogous to the barbell strategy in fixed income.

Warfare Economics Convexity
The premium in military effectiveness achieved by temporally dispersing resource commitments (R&D, procurement, operations) rather than concentrating them.

Graph Neural Network (GNN) Message Passing
A machine learning mechanism where node representations are updated by aggregating neighbor information, structurally isomorphic to recursive DACPT backward induction.

Fractal Pricing Theory
A financial framework where the same pricing principles (duration anchor + convexity adjustment) apply identically at micro, meso, and macro scales.

The End